have become quite popular, especially with the two international robot sporting competitions – RoboCup and MIROSOT. Several robots need to work individually as well as in tandem with one another. This is possible thanks to a branch of AI called Distributed Artificial Intelligence – the same kind of software that powers web-bots or know-bots.

7. **Nanobots:** Robots at the nanometre level are still more sci-fi than reality, but recent advances in nanotechnology have made them a distinct possibility sometime in the near future.

How robots work

In popular science fiction, Isaac Asimov's three laws of robotics describe how robots should be programmed:

- A robot may not injure a human being or, through inaction, allow a human being to come to harm.
- A robot must obey any orders given to it by human beings, except where such orders would conflict with the First Law.
- A robot must protect its own existence as long as such protection does not conflict with the First or Second Law.

However, robot programmers haven't really reached far ahead yet in reality.

Face forward

In May 2010, Microsoft released its first stable version of the Microsoft Robotics Developer Studio (Microsoft RDS, MRDS) - a Windows-based environment for robot control and simulation. It has a high degree of compatibility and is useful for students, professionals and commercial applications as well. The interest of giants such as Microsoft in robotics signals an exciting future

Spielberg's version of I, Robot revealed a fallacy in the three laws

in the field. Panasonic's HOSPI (a hospital robot that serves, cleans and takes X-rays), humanoids like TOPIO (a robot that plays table tennis) and recent bionic innovations indicate that a growing presence of robots in our daily life is inevitable. Automated transport vehicles, rescue robots and robotic explorers are already being worked on to be enhanced for interaction with the general populace. The multi-disciplinary influences of cognitive sciences, bio-robotics and linguistics are bound to come up with better robots. So, although you're not going to see WALL-E just yet, you will see robots help you out with your housework in the near future.

India connections

According to Prof. Gayatri Menon, Faculty and Coordinator of the Toy and Game Design Department at the National Institute of Design, Gandhinagar, industrial robotics have seen a spurt in development in India over the last decade. ABB, the European automation giant, has a burgeoning business in India, with Indian companies buying industrial robots for a variety of functions like welding, paint finishing and material handling. In view of this, it has set up a Robotics Application Centre in Bangalore especially for the purpose. Prof. Menon feels that the Indian industries have shed their inhibitions about using robots in their factories over the last few years and the perception that roboticization would lead to unemployment is no longer in vogue.

Start-ups like Pulkit Gaur's Gridbots (Ahmedabad), Dr. Prahlad Vedakkepat's Robhatah (Bangalore) and Abheek Bose's Robots Alive (Bangalore) are not only actively pursuing innovations in robotics but also interacting with faculty, students and researchers through specially designed workshops on robotic technology. These are usually designed to provide a balanced exposure to the theoretical as well as the practical aspects to robotics.

However, Prof. Menon feels that the introduction of Mechatronics as a subject in engineering colleges in India – and the growing number of robotics competitions such as the SmartMove contest in Bangalore on January 30, 2010 – are steps in the right direction and will serve to increase the awareness of the layman to robotics.

Pranjal Rai, a research student in Education through robotics at the National Institute of Design, Ahmedabad, who has spent several months surveying the robotics scene in schools of India, found that most school authorities and faculty see robotics as a hobby activity. At the personal level, Lego's Mindstorms kits took the world by storm in 1998 but are still quite expensive by Indian standards. Rai, to his surprise, found that most robot-kits for children ended up in disuse, being ignored after a few attempts at play. In fact, let's face it, for most of us, studying robotics

Prof. Gayatri Menon sees a bright future for Robotics in Education

is not just a geek-thing, but perhaps the geekiest of all the geeky things to do.

The general perception in India is that robotics is hyper-technical. Prof. Menon believes that a serious attempt needs to be made to dispel these misgivings – because robotics is probably the only thing that could teach children mechatronics, computer programming, electronics, mechanics and bio-mechanics in one shot – and with tons of fun in the bargain. No doubt, if India wants to seriously enter the international robotics scene in a significant way and make good use of the vast pool of technical talent that we already have a collaborative effort between industry and academia is the need of the hour. Watch out for next month's Collector's Edition for more on robotics.